Discrete Math Course Notes

Stephen Arndt
Instructor: Florian Frick

Fall 2024

1 Probabilistic Combinatorics

08/26 - 08/30 Notes: Hypergraphs

Definition 1.1. A c**-coloring** of a hypergraph H is a function $f : V \to \{1, 2, \ldots, c\}$ s.t. every edge $e \in H$ has at least two colors.

Let $m(k)$ be the **smallest** number of edges of a k-uniform hypergraph H that does **not** admit a 2-coloring.

Theorem 1.2. *$m(k) \geq 2^{k-1}$ for all $k \geq 1$.*

Proof. Let H be a k-uniform hypergraph s.t. $|H| < 2^{k-1}$. Pick a random 2-coloring. For each edge $e \in H$, the probability e is red or blue is $1/2^{k-1}$. Thus the probability some $e \in H$ is red or blue is $\leq |H|/2^{k-1} < 1$. Thus H has a 2-coloring. □

Theorem 1.3. *Let $G = (V, E)$ be an undirected graph. Then G has a cut $(S, V \backslash S)$ with $\geq m/2$ edges.*

Proof. Let $x_1, x_2, \ldots, x_n$ be independent 0/1 r.v.'s with $p = 1/2$. Let $S = \{i : x_i = 1\}$. An edge $(i, j) \in E$ is in the cut iff $x_i \neq x_j$. Thus expected number of edges in cut is $m/2$. Thus there is a cut with $\geq m/2$ edges. □

Remark 1.4. Theorem 1.3 is almost optimal. Use $G = K_n$. $m \approx n^2/2$. Max number of cut edges is $(n/2)(n/2) = n^2/4$.

08/30 - 09/04 Notes: Antichains and Intersecting Sets

Definition 1.5. $\mathcal{F} \subseteq 2^{[n]}$ is an **antichain** if for all distinct $A, B \in \mathcal{F}$, $A \nsubseteq B$, $B \nsubseteq A$.

Definition 1.6. $\mathcal{C} \subseteq 2^{[n]}$ is a **chain** if for a permutation $\sigma : [n] \to [n]$, $\mathcal{C}$ is totally ordered by inclusion according to σ.

Fact 1.7. *An antichain $\mathcal{F}$ intersects a chain $\mathcal{C}$ at most once.*

Theorem 1.8 (Sperner's Theorem). *The largest antichain of* $[n]$ *has size* $\binom{n}{n/2}$.

Lemma 1.9 (LYM-Inequality). *Let* $\mathcal{F} \subseteq 2^{[n]}$ *be an antichain. Then*

$$\sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}} \leq 1$$

Proof. Pick a random chain $\mathcal{C}$. For each $A \in \mathcal{F}$, let $X_A = 1$ if $A \in \mathcal{C}$, 0 else. $\Pr[A \in \mathcal{C}] = \frac{1}{\binom{n}{|A|}}$. Thus

$$\mathbb{E}[\mathcal{F} \cap \mathcal{C}] = \sum_{A \in \mathcal{F}} \mathbb{E}[X_A] = \sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}} \leq 1$$

□

Definition 1.10. $\mathcal{F} \subseteq 2^{[n]}$ is **intersecting** if for all $A, B \in \mathcal{F}$, $A \cap B \neq \emptyset$.

Theorem 1.11 (Erdös-Ko-Rado Theorem). *Let* H *be an intersecting* k*-uniform hypergraph with* $n \geq 2k$. *Then* $|H| \leq \binom{n-1}{k-1}$.

Proof. First, represent $\mathbb{Z}/n$ as a cycle on n vertices. Let $A_s = \{s, s+1, \ldots, s+k-1\} \subseteq \mathbb{Z}/n$ for each $s = 0, 1, \ldots, n-1$. I claim the largest intersecting subfamily of $\{A_s : s \in \mathbb{Z}/n\}$ has size $\leq k$. Let $\mathcal{F} \subseteq \{A_s : s \in \mathbb{Z}/n\}$ be intersecting. Then the union of $\mathcal{F}$ must form a single path $P = (u, u+1, \ldots, v)$ on the cycle, or else some $A_i, A_j \in \mathcal{F}$ are disjoint. P must have length at most $2k-1$, or else $A_u, A_{u+k} \in \mathcal{F}$ are disjoint. Thus $|\mathcal{F}| \leq k$.

Next, consider random sampling from $\Omega = S_n \times \mathbb{Z}/n$. If we uniformly sample (π, s), $\pi(A_s)$ is a uniform random k-subset, so $\Pr[\pi(A_s) \in H] = |H|/\binom{n}{k}$. For all fixed π, if we uniformly sample s, then $\Pr[\pi(A_s) \in H] \leq k/n$. Thus $|H|/\binom{n}{k} \leq k/n$ giving $|H| \leq \binom{n-1}{k-1}$. □

Conceptual Idea: We can perform **global sampling** and **local sampling.** If we want to show some property of a combinatorial object holds under "global" sampling (ex. a bound), we can show this property holds under "local" sampling (where some r.v.'s might get fixed).

09/04 - 09/06 Notes: Alterations

Alterations

- Probabilistic technique for getting "large" combinatorial object
- Sample X things to get a random object
- Remove Y things to get a cleaned object

Theorem 1.12. *For all* $n \in \mathbb{N}$, *there exists a graph* G *with...*

- ***Property:*** $K_{3,3}$*-free*

- ***Size:*** $m = \Omega(n^{3/2})$ *edges*

Proof. Sample $G = G(n,p)$. Let X be number of edges. Let Y be number of $K_{3,3}$ subgraphs, defined by $A, B \in \binom{[n]}{3}$. Remove one edge from each $K_{3,3}$ subgraph. G has $X - Y$ edges. Compute $\mathbb{E}[X - Y]$ and pick $p = \Theta(n^{-1/2})$. $\square$

Theorem 1.13. *Let G be a graph on n vertices with average degree $d \geq 1$. Then $\alpha(G) \geq n/2d$.*

Proof. Sample a random subset $S \subseteq V$ by including each vertex independently with probability p. Let $X = |S|$ and $Y = |E(G[S])|$. Remove one vertex $x \in X$ from each edge $y \in Y$. S has $X - Y$ vertices. Compute $\mathbb{E}[X - Y]$ and pick $p = 1/d$. $\square$

Definition 1.14. For a graph G, the **girth** $g(G)$ is the length of a **shortest cycle**. If G has no cycles, $g(G) = +\infty$.

Definition 1.15. For a graph G, the **chromatic number** $\chi(G)$ is the **minimum** number of **colors** needed to color V.

Conceptually, it seems high girth $\implies$ low density, high chromatic number $\implies$ high density.

Theorem 1.16 (Erdös)**.** *For every $k, \ell \geq 0$, there is a graph G with $g(G) \geq \ell$, $\chi(G) \geq k$.*

Proof Strategy. Sample $G = G(n,p)$. Let X be the number of short cycles (length $\leq \ell - 1$) in G. We want $\mathbb{E}[X] = o(n)$ and $\alpha(G) = o(n)$ (with high probability). If so, we can perform alterations. Remove one vertex from each short cycle. Note this only decreases $\alpha(G)$. This leaves G with expected $n - o(n)$ vertices (importantly, G is nonempty with high probability), $g(G) \geq \ell$, and $\chi(G) \geq \frac{n}{\alpha(G)} \to \infty$.

Proof Details. $\mathbb{E}[X] \leq \sum_{t=3}^{\ell-1} n^t p^t$, so we consider $p = n^{-1+\epsilon}, \epsilon > 0$. $\epsilon = \frac{1}{2\ell}$ works. Now the original G is completely well-defined, so we just do analysis to bound $\alpha(G)$. We upper bound $\Pr[\alpha(G) \geq a]$ where a is a parameter, and pick a to make sure $a = o(n)$ and $\Pr[\alpha(G) \geq a] \to 0$ as $n \to \infty$. $\square$

09/09 - 09/11 Notes: Second Moment Method

Second Moment Method: The study of variance. Let X be a random variable. If $\mathrm{Var}[X]$ is low, then X is tight around $\mathbb{E}[X]$. If $\mathbb{E}[X]$ is large, then we can argue $\Pr[X = 0]$ is low.

Markov vs. Chebyshev Inequalities: Let X be a random variable, and consider its distribution. Markov's Inequality bounds the **right tail** of the distribution. Chebyshev's Inequality bounds **both tails** of the distribution. Thus Chebyshev's Inequality allows us to bound the **left tail** of the distribution, and argue that the probability X is very low (commonly $X = 0$) is very low.

Definition 1.17. Let X be a random variable. Then $\mathrm{Var}[X] = \mathbb{E}[(X-\mu)^2]$.

Definition 1.18. Let X, Y be random variables. Then
$\mathrm{Cov}[X,Y] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

Lemma 1.19 (Variance and Covariance Facts)**.**

- $Var[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.
- $Var(\sum X_i) = \sum Var(X_i) + 2 \cdot \sum_{i<j} Cov(X_i, X_j)$.
- $Cov(X_i, X_j) = 0$ *if* X_i, X_j *independent.*

Lemma 1.20 (Chebyshev's Inequality)**.** *Let X be a random variable. Then for all $t \geq 0$,*

$$\Pr[|X - \mathbb{E}[X]| \geq t] \leq \frac{Var(X)}{t^2}$$

Consider $t = \mathbb{E}[X]$.

Corollary 1.21. *Let X be a random variable. Then* $\Pr[X = 0] \leq \frac{Var(X)}{\mathbb{E}[X]^2}$.

We now consider an example problem.

Theorem 1.22. *Let $G = G(n,p)$. Let X be the number of triangles in G. If $np \to \infty$, then* $\Pr[X=0] \to 0$.

Proof. Write $X = \sum_{\tau \in \binom{[n]}{3}} X_\tau$ where $X_\tau = 1$ if τ is a triangle in G, 0 else. Use $\Pr[X=0] \leq \frac{\mathrm{Var}(X)}{\mathbb{E}[X]^2}$ and do computations to get the result. □

Threshold Phenomenon: Suppose $G \sim G(n,p)$.

- $\Pr[G$ has triangle$] \to 0$ for $p \ll 1/n$. (Markov's Inequality)
- $\Pr[G$ has triangle$] \to 1$ for $p \gg 1/n$. (Chebyshev's Inequality)

Definition 1.23. Let G be a graph. The **density** of G is $\delta(G) = m/n$. G is **balanced** if no subgraph H of G has higher density than G.

Theorem 1.24. *Let H be a balanced graph. Then the threshold function for the appearance of H in $G = G(n,p)$ is $p = n^{-1/\delta(H)}$.*

Proof Idea. Suppose H has v vertices and e edges. For an ordered v-tuple $\alpha = (a_1, a_2, \ldots, a_v)$ of distinct vertices in G, let $X_\alpha = 1$ if H is a subgraph of $G[\alpha]$ in the "specific order" given by α, 0 otherwise. Let $X = \sum X_\alpha$. Do computations to get the result. □

Let $f(n)$ be the maximum k for which there is a set $\{x_1, x_2, \ldots, x_k\} \subseteq [n]$ with distinct sums in $[n]$.

Theorem 1.25. $f(n) \leq (1 + o(1)) \log_2 n$.

Proof Idea. Let $\{x_1, x_2, \ldots, x_k\} \subseteq [n]$ have distinct sums. Let $\epsilon_1, \epsilon_2, \ldots, \epsilon_k$ be independent 0/1 r.v.'s. with $p = 1/2$. Consider $X = \sum \epsilon_i x_i$. Via Chebyshev's Inequality, X is tight around $\mathbb{E}[X]$. Via the Distinct Sums condition, X is loose around $\mathbb{E}[X]$. Formally stating these observations leads to n large w.r.t. k. $\square$

Remark 1.26. Theorem 1.25 easily solved by Pigeonhole Principle. There are $2^k - 1$ nonempty subsets which must have distinct sums in $[n]$.

09/13 - 09/18 Notes: Lovász Local Lemma

Suppose we have (Ω, p) where $\Omega =$ outcomes, $p =$ probability.

Definition 1.27. $A, B \subseteq \Omega$ are **independent** if $p(A \cap B) = p(A)p(B)$.

Definition 1.28. Let $A, B_1, \ldots, B_k \subseteq \Omega$ be events. Then A is **mutually independent** of $B_1, B_2, \ldots, B_k$ if for all nonempty $J \subseteq [k]$,

$$p\left(A \cap \left(\bigcap_{i \in J} B_j\right)\right) = p(A) \cdot p\left(\bigcap_{i \in J} B_j\right)$$

If a set of events is referred to as "independent", this typically means that they are mutually independent. Further, mutual independence is **stronger** than pairwise independence between all $A, B_1, B_2, \ldots, B_k$! Often we need **mutual independence** to make simplifications.

Fact 1.29.

$$p\left(\bigcap_{i=1}^{n} \bar{A}_i\right) = p\left(\bar{A}_1\right) \cdot p\left(\bar{A}_2 | \bar{A}_1\right) \cdot p\left(\bar{A}_3 | \bar{A}_1 \cap \bar{A}_2\right) \cdot \cdots \cdot p\left(\bar{A}_n | \bar{A}_1 \cap \bar{A}_2 \cap \cdots \cap A_{n-1}^{-}\right)$$

Conceptual Idea: Useful in small cases. If $A_1, A_2, \ldots, A_n$ are "bad" events, this allows for a direct computation of the probability that no bad event occurs.

Definition 1.30. Let $A_1, A_2, \ldots, A_n \subseteq \Omega$ be events. A **dependency digraph** is a directed graph D on $[n]$ s.t. A_i is mutually independent of $\{A_j | (i, j) \notin E(D)\}$ for each $i \in [n]$.

Lemma 1.31 (Lovász Local Lemma)**.** *Let $A_1, A_2, \ldots, A_n \subseteq \Omega$ be events with dependency digraph $D = ([n], E)$. If there exists $x_1, x_2, \ldots, x_n \in [0, 1)$ s.t. $p(A_i) \leq x_i \cdot \prod_{(i,j) \in E} (1 - x_j)$ for each $i \in [n]$, then $p\left(\cap_{i=1}^{n} \bar{A}_i\right) \geq \prod_{i=1}^{n} (1 - x_j) > 0$.*

Lemma 1.32 (Symmetric LLL)**.** *Let $A_1, A_2, \ldots, A_n \subseteq \Omega$ be events with dependency digraph $D = ([n], E)$. Suppose $p(A_i) \leq p$ for all $i \in [n]$, and the outdegrees in D are at most d. If $e \cdot p(d + 1) \leq 1$, then $p\left(\cap_{i=1}^{n} \bar{A}_i\right) > 0$.*

Conceptual Idea: Let $A_1, A_2, \ldots, A_n$ be "bad" events. Suppose each event A_i occurs with **low probability** and has **low dependency** on the other events.

Then $p\left(\cap_{i=1}^n \bar{A}_i\right) > 0$, meaning with nonzero probability, **none** of the **bad events** occur.

Note: It's common to consider D as an undirected graph. This doesn't change the statement of the lemmas.

Problem 1.33 (RIS Problem)**.** Let C_n be an n-cycle, where $n = c \cdot k$. Suppose the vertices are split into c color classes of size k. What is the minimum value of k so that there exists a **rainbow independent set?**

Solution 1.34. Uniformly pick a vertex of each color to construct S. S is a rainbow set. Bad events $A_i = \{S \mid i, i+1 \in S\}$ for $i \in [n]$ (where $i+1$ is mod n). $p(A_i) = 1/k^2$ if $i, i+1$ different colors, 0 if $i, i+1$ same color for each $i \in [n]$. For dependency, suppose you know some event A_i occurs where $i \in [n]$. Suppose i is red and $i+1$ is blue. This affects all other events which involve a red vertex or a blue vertex, of which there are $2(k-1)$ each; and the 2 adjacent events A_{i-1} and A_{i+1}. Thus the dependency $d \leq 2(k-1)+2(k-1)+2 = 4k-2$. Rearranging $e \cdot p(d+1) \leq 1$ yields $k \geq 11$. Thus $k = 11$ works.

Dependency Analysis: To figure out dependencies, it seems useful to suppose an event occurs, and think of objects or events which have **lost independence** as being **"corrupted"**.

Given a coloring $c : \mathbb{R} \to [k]$, a set $A \subseteq \mathbb{R}$ is **colorful** if $c(A) = [k]$.

Theorem 1.35. *$\forall k \geq 1, \exists m \in \mathbb{N}$ s.t. $\forall S \subseteq \mathbb{R}$ with $|S| = m$, there exists a coloring $c : \mathbb{R} \to [k]$ s.t. $x + S$ is **colorful** for all $x \in \mathbb{R}$.*

Lemma 1.36 (Finite Version of Theorem 1.35)**.** *Same as Theorem 1.35, except we claim $x + S$ is **colorful** for all $x \in X$ where $X \subseteq \mathbb{R}$ is a finite set.*

Proof. Pick c random by coloring each $y \in \cup_{x \in X}(x + S)$ randomly. Bad events are $A_x = \{c \mid x + S \text{ is not colorful}\}$. $\Pr[A_x] = \cup_{i \in [k]}(x + S \text{ is missing color } i) \leq k(1 - 1/k)^m \leq ke^{-m/k}$ for all $x \in X$. Thus $p = O(\exp(-m))$. For dependency, each $x + S$ intersects at most $m(m-1) = d$ other translates $y + S$, because the i'th element of $x + S$ could be the j'th element of $y + S$ for each distinct $i, j \in [m]$. Thus $d = O(m^2)$. Thus $ep(d+1) = o(1)$, proving the claim. $\square$

My vague understanding of "compactness" is that it allows us to claim that an infinite counterexample must contain a finite counterexample. Thus Theorem 1.35 holds via Lemma 1.36.

09/20 - 09/23 Notes: Chernoff Bounds

Theorem 1.37 (Chernoff Bound)**.** *Let $X_1, X_2, \ldots, X_n$ be independent r.v.'s with $\mathbb{E}[X_i] = 0$ and $|X_i| \leq 1$ for each $i \in [n]$. Then $\Pr\left[\sum X_i \geq t\right] \leq e^{-t^2/(2n)}$ for all $t \geq 0$.*

Conceptual Idea: If $X_1, X_2, \ldots, X_n$ are **independent** r.v.'s, their sum $X = \sum X_i$ is **very tight** around its mean $\mathbb{E}[X]$. Specifically, Chernoff Bounds give **exponential** tightness, while Chebyshev gives **quadratic** tightness.

Note: We really only require that X_i's bounded. If so, we can normalize and apply Chernoff Bound.

Theorem 1.38 (Hamming Distance in Boolean Hypercube)**.** *Let $\delta \in (0, 1/2)$. Pick two points $X, Y \in \{0,1\}^n$ uniformly at random. Define $d_H(X,Y)$ as the number of bit flips between X and Y. Then* $\Pr[d_H(X,Y) \leq \delta n] \leq \exp\left(\frac{(1-2\delta)^2 n}{2}\right)$.

Proof. Let $Z_i = 1$ if X, Y differ at position i, 0 else for each $i \in [n]$. Let $Z = \sum Z_i$. We want to bound $\Pr[Z \leq \delta n]$. Let $\hat{Z}_i = 1 - 2Z_i$ for each $i \in [n]$ for normalization. $\mathbb{E}[\hat{Z}_i] = 0$, $|\hat{Z}_i| \leq 1$ for each $i \in [n]$. $\hat{Z} = \sum \hat{Z}_i$. Express $\Pr[Z \leq \delta n]$ using $\hat{Z}$ and apply Chernoff Bound to get result. ☐

Lemma 1.39. *Let $Y_1, Y_2, \ldots, Y_n$ be independent r.v.'s. Then* $\mathbb{E}\left[\prod Y_i\right] = \prod \mathbb{E}[Y_i]$.

Proof Idea. First, prove for indicator random variables. Suppose $Y_i = 1$ if event A_i occurs, 0 else for each $i \in [n]$. $\mathbb{E}\left[\prod Y_i\right] = p(\cap A_i) = \prod p(A_i) = \prod \mathbb{E}[Y_i]$. Next, recall r.v.'s are functions from Ω to $\mathbb{R}$, and indicator r.v.'s are functions from Ω to $\{0,1\}$. Thus all r.v.'s are **linear combinations** of indicator r.v.'s. Using linearity of expectation, because the identity holds for independent indicator r.v.'s, it holds for general independent r.v.'s. ☐

Lemma 1.40. *Let G be a 1000-regular graph. Then there exists a subset of vertices $U \subseteq V$ s.t. in $G[U]$, all degrees are between 100 and 900.*

Proof. A natural initial strategy is Chernoff Bounds. Pick a random subset $U \subseteq V$ by including each vertex independently with probability $p = 1/2$. Let $A_v = \{U \subseteq V : v \in U \text{ and } (d(v) \leq 100) \text{ or } (d(v) \geq 900)\}$ for each $v \in V$. The A_v's are the "bad events." Using Chernoff Bounds, we can conclude $p(A_v)$ is low for each $v \in V$. Crucially though, it is constant. Thus we cannot union bound over all $v \in V$, because $|V| = n$...!

Instead, we should use Symmetric LLL. We can bound $p(A_v) \lessapprox e^{-80}$ for each $v \in V$. For dependency, suppose event A_v occurs for some $v \in V$. Then all of v's neighbor nodes $w \in V$ are "corrupted." Then **any** event which uses a neighbor node $w \in V$ is "corrupted." There are $\leq 1000^2$ such events, given by the neighbors of all $w \in V$. Using $p \approx e^{-80}$ and $d = 1000^2$, we can verify $ep(d+1) \leq 1$. Thus with nonzero probability, no bad event A_v occurs. Thus there exists a subset $U \subseteq V$ s.t. $G[U]$ has all degrees between 100 and 900. ☐

Conceptual Idea: A **major strength** of Lovász Local Lemma is that the number of bad events can be **unbounded.** Union Bounds typically required the number of bad events to be **bounded.**

Definition 1.41. A sequence $X_1, X_2, \ldots, X_n$ of r.v.'s is a **martingale** if $\mathbb{E}[X_{i+1} | X_1, X_2, \ldots, X_i] = X_i$ for each $i \in [n-1]$.

Theorem 1.42 (Azuma's Inequality)**.** *Let $X_1, X_2, \ldots, X_n$ be a martingale with $\mathbb{E}[X_n] = 0$ and $|X_{i+1} - X_i| \leq 1$ for each $i \in [n-1]$. Then $\Pr[|X_n| \geq t] \leq e^{-t^2/2n}$.*

Conceptual Idea: If $X_1, X_2, \ldots, X_n$ is a martingale with expected ending value of 0, and each X_i is close to the previous X_{i-1}, then X_n will be very tight around 0.

2 Extremal Combinatorics

09/23 - 09/25 Notes: Ramsey Theory

Definition 2.1. The **Ramsey number** $R(s,t)$ is the minimum number n s.t. all n-vertex graphs G have an **independent set** of size $\geq s$ or a **clique** of size $\geq t$.

Definition 2.2 (Equivalent Definition)**.** The **Ramsey number** $R(s,t)$ is the minimum number n s.t. for all **red/blue colorings** of the edges of K_n, there is a **red K_s-subgraph** or a **blue K_t-subgraph.**

Theorem 2.3. *$R(t,t) \geq 2^{t/2} + 1$ for $t \geq 3$.*

Proof. Probabilistic Method. Let $n = 2^{t/2}$. Randomly sample a 2-coloring. For each $T \subseteq V$ s.t. $|T| = t$, compute probability that $K_n[T]$ monochromatic. Union bound over all such T. With probability < 1, there is a monochromatic t-clique. With probability > 0, there is no monochromatic t-clique. □

Theorem 2.4. *$R(t,t) \leq 2^{2t}$.*

Proof. Induction and PHP. Let $c : K_{2^{2t}} \to \{R, B\}$ be arbitrary. We will show the existence of a monochromatic t-clique. To do so, we will construct a sequence of vertices $v_1, v_2, \ldots, v_{2t}$ with strong monochromatic properties. First, arrange the vertices $1, 2, \ldots, 2^{2t}$ on a line (useful for visualization). Let $v_1 = 1$. Consider all edges coming out of 1. Let S_1 be the set of vertices from $2, 3, \ldots, 2^{2t}$ which has the "majority" edge color from 1. Let $v_2 = \min(S_1)$. Consider all "forward" edges coming out of v_2. Let S_2 be the set of vertices from $v_2 + 1, v_2 + 2, \ldots, 2^{2t}$ with the "majority" edge color from v_2. Let $v_3 = \min(S_2)$. Repeat this procedure until we obtain $v_1, v_2, \ldots, v_{2t}$ and $S_1 \supseteq S_2 \supseteq \cdots \supseteq S_{2t}$. A key observation is that on each step, we keep at least half of the remaining vertices. Thus we will actually be able to make this sequence length $2t$. Finally, each v_i has all forward edges monochromatic to $v_{i+1}, v_{i+2}, \ldots, v_{2t}$. At least t vertices have the same forward edge color. These vertices form a monochromatic t-clique. □

Conceptual Strategy: This strategy is motivated by the following **major limitation.** The coloring is fixed and arbitrary, so we can only use **Pigeonhole Principle** to get monochromatic structures. Further, PHP can only say the **weak condition** that among a set of edges, at least half are the same color. Thus, to get a complex monochromatic structure, we could start with 1, and make the basic observation that at least half of 1's edges are monochromatic.

From there, we need to stay working with the identified vertices, because we have no grip on the other vertices (there could be $\approx$ half of them, or very few of them). Continuing this reasoning leads to the iterative argument above.

In future theorems with arbitrary colorings, we will see the same **major limitation** motivate the same **iterative argument!**

Theorem 2.5 (Hypergraph Ramsey Theorem ($k = 2$))**.** *For any red/blue coloring of $\binom{\mathbb{N}}{2}$, there is an infinite set $S \subseteq \mathbb{N}$ s.t. all 2-subsets of S are the same color.*

Proof. Let $c : \binom{\mathbb{N}}{2} \to \{R, B\}$ be arbitrary. Same iterative argument as Theorem 2.4. We get a sequence $v_1, v_2, \dots$ and $S_1 \supseteq S_2 \supseteq \dots$. The key observation, which allows propagation, is that given v_i and its forward edges, an **infinite set** of those forward edges are the same color. $\square$

Theorem 2.6 (Hypergraph Ramsey Theorem)**.** *For any red/blue coloring of $\binom{\mathbb{N}}{k}$, there is an infinite set $S \subseteq \mathbb{N}$ s.t. all k-subsets of S are the same color.*

Proof. Let $c : \binom{\mathbb{N}}{k} \to \{R, B\}$ be arbitrary. We use the same iterative argument. Start with $v_1 = 1$. On a basic level, we know an infinite number of k-subsets including 1 are the same color. However, this does not give us what we want. We want an infinite subset S_1 s.t. all k-subsets of S_1 including 1 are the same color. We can see this comes from induction. Consider the coloring c' of $(k-1)$-subsets from $2, 3, \dots$ induced by the coloring c on k-subsets including 1. By induction using c', get an infinite monochromatic subset $S_1 \subseteq \{2, 3, \dots\}$. Now we know how to propagate the sequence $v_1, v_2, \dots$ and $S_1 \supseteq S_2 \supseteq \dots$. $\square$

Note: Generalizes to j-colorings for $j \geq 2$! The argument is identical. For propagation, we can still get infinite monochromatic subsets S_i.

09/27 - 10/02 Notes: Extensions (EST, VWT, HJT)

Definition 2.7. $X \subseteq \mathbb{R}^d$ is in **convex position** if no $x \in X$ is in the convex hull of $X \backslash \{x\}$. $X \subseteq \mathbb{R}^d$ is in **general position** if no three points in X are collinear.

Theorem 2.8 (Erdös-Szekeres Theorem)**.** *$\forall k \in \mathbb{N}$, $\exists n \in \mathbb{N}$ s.t. for all $X \subseteq \mathbb{R}^2$ in general position with $|X| = n$, there exists a subset $Y \subseteq X$ in convex position with $|Y| = k$.*

Proof. Given $k \in \mathbb{N}$, let $n \in \mathbb{N}$ be arbitrary. A key observation is that k points are in convex position $\iff$ any 4 of them are in convex position. Define $c : \binom{[n]}{4} \to \{R, B\}$ s.t. $c(S) =$ blue is S in convex position, red otherwise for each $S \in \binom{[n]}{4}$. If n is sufficently large, there is a red 5-clique or a blue k-clique. A red 5-clique is impossible, so we must get a blue k-clique. $\square$

Theorem 2.9 (Van der Waerden's Theorem ($m = 3$))**.** *For any k-coloring of $\mathbb{N}$, there exists a monochromatic 3-AP.*

Proof. Let $c : \mathbb{N} \to [k]$ be arbitrary. A set of monochromatic 2-APs is **color-focused** if a) the 2-APs are distinct colors, and b) the 2-APs coincide at the same "focus" $f \in \mathbb{N}$ when extended to 3-APs. A key observation is that a set of k monochromatic color-focused 2-APs yields a monochromatic 3-AP. Thus the following proposition proves Theorem 2.9.

Proposition 2.10. *For all $r \leq k$, there exists an $n \in \mathbb{N}$ s.t. whenever n is k-colored, i) there exists a monochromatic 3-AP or ii) there are r color-focused 2-APs.*

Using induction on r, the core argument depends on showing how to extend $r-1$ color-focused 2-APs to r color-focused 2-APs. Let $n \in \mathbb{N}$ be the number s.t. in any block of size n, there are $r-1$ color-focused 2-APs. A key observation is that, given a coloring of $c : \mathbb{N} \to [k]$, entire blocks of size $2n$ (or any finite size) **must** repeat. If we find two blocks B_s, B_{s+t} which are identical, and in which we know there are $r-1$ identical color-focused 2-APs (with foci inside the blocks), we can thread the $r-1$ 2-APs together, and get another 2-AP from the foci. This completes the induction. □

Conceptual Strategy: This strategy is motivated by the same **major limitation** as Theorem 2.4. First, the idea of considering color-focused 2-APs is nontrivial, but it makes sense how the idea allows us to keep "adding pressure" onto the coloring, until a 3-AP is pushed out. Next, we prove the proposition. The coloring is fixed and arbitrary, so we can only use a) the inductive hypothesis, and b) in infinite settings, even **big stuff** has **repeats** (PHP). We then give ourselves the freedom to extremely abuse the second observation.

Theorem 2.11 (Van der Waerden's Theorem)**.** *$\forall m \in \mathbb{N}$, for any k-coloring of $\mathbb{N}$, there exists a monochromatic m-AP.*

Proof. Induction on m. Use same proposition as Theorem 2.9. For the proposition, to get the extra color-focused $(m-1)$-AP, we extremely abuse the global inductive hypothesis. Get a "monochromatic" $(m-1)$-AP of blocks $B_s, B_{s+t}, \ldots, B_{s+(m-2)t}$ of size $2n$, where "block colors" are defined by k-coloring in the block, so that there are k^{2n} "block colors." Thread the $r-1$ monochromatic color-focused $(m-1)$-APs in each block together, and thread the $m-1$ foci together, to get r monochromatic color-focused $(m-1)$-APs. □

Conceptual Strategy: Same as before, but we give ourselves the freedom to extremely abuse the global inductive hypothesis.

Definition 2.12. A **combinatorial line** $L \subseteq [m]^n$ is defined by a nonempty subset of indices $I \subseteq [n]$, where L is the set of points $(x_1, x_2, \ldots, x_n) \in [m]^n$ s.t. $x_i = x_j$ for all $i, j \in I$, and $x_i = a_i$ for all $i \in [n] \backslash I$ where the a_i's are constant.

Theorem 2.13 (Hales-Jewett Theorem)**.** *$\forall m, k \in \mathbb{N}$, $\exists n \in \mathbb{N}$ s.t. for any k-coloring of $[m]^n$, there is a monochromatic combinatorial line.*

Proof. Induction on m. The **"last" point** on a combinatorial line L is given by setting $x_i = m$ for all $i \in I$. A combinatorial line is **almost-monochromatic** if it is monochromatic, except possibly its last point. A set of almost-monochromatic combinatorial lines is **color-focused** if a) the combinatorial lines are distinct colors, and b) the last points on each combinatorial line coincide.

Proposition 2.14. *For all $r \leq k$, there exists an $n \in \mathbb{N}$ s.t. whenever $[m]^n$ is k-colored, i) there is a monochromatic combinatorial line or ii) there are r color-focused combinatorial lines.*

Induction on r. Suppose $n \in \mathbb{N}$ gives $r-1$ almost-monochromatic color-focused combinatorial lines. Now we extremely abuse both inductive hypotheses. Consider $[m]^n$ as a "block" of one color in $[k^{m^n}]$. Consider putting these $[m]^n$ blocks into an $[m]^{n'}$ lattice. Pick n' large enough so that there exists a length $m-1$ monochromatic combinatorial line of blocks. These blocks are identical, and each contain $r-1$ almost-monochromatic color-focused combinatorial lines. Thread these $r-1$ color-focused combinatorial lines in each block together, and thread the $m-1$ foci together, to get r almost-monochromatic color-focused combinatorial lines. $\square$

10/04 - 10/09 Notes: Ultrafilters

Definition 2.15. $\mathcal{U} \subseteq 2^{\mathbb{N}}$ is an **ultrafilter** on $\mathbb{N}$ if...

1. **Closed under Intersection:** $A, B \in \mathcal{U} \implies A \cap B \in \mathcal{U}$.
2. **Closed under Supersetting:** $A \in \mathcal{U}, B \supseteq A \implies B \in \mathcal{U}$.
3. **Partitioning:** For any partition $\mathbb{N} = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_k$, **exactly one** $A_j \in \mathcal{U}$.

Equivalent Condition 3 ($k = 2$): For all $A \subseteq \mathbb{N}$, exactly one of $A, A^c \in \mathcal{U}$.

There are two types of ultrafilters. An ultrafilter is **principal** if it contains a finite set, and is **free** if it contains only infinite sets. We can see that principal ultrafilters always take the form $\mathcal{U} = \{X \cup \{x\} : X \subseteq \mathbb{N} \backslash \{x\}\}$ for some $x \in \mathbb{N}$.

Definition 2.16. $\mathcal{F} \subseteq 2^{\mathbb{N}}$ is a **filter** on $\mathbb{N}$ if $\emptyset \notin F$, and...

1. **Closed under Intersection:** $A, B \in \mathcal{F} \implies A \cap B \in \mathcal{F}$.
2. **Closed under Supersetting:** $A \in \mathcal{F}, B \supseteq A \implies B \in \mathcal{F}$.

Definition 2.17 (Equivalent Definition of Ultrafilter)**.** $\mathcal{U} \subseteq 2^{\mathbb{N}}$ is an **ultrafilter** on $\mathbb{N}$ if it is an **inclusion-maximal filter.**

We now give some definitions to build up to Zorn's Lemma. Let P be a poset, where P is a collection of subsets ordered by inclusion.

Definition 2.18. $\{A_\alpha | \alpha \in J\}$ is a **chain** if it is a totally ordered set.

Definition 2.19. B is an **upper bound** of chain $\{A_\alpha : \alpha \in J\}$ if B is in poset, and $A_\alpha \subseteq B$ for all $\alpha \in J$.

Definition 2.20. M is a **maximal element** of P if for all $N \in P$, $N \not\supseteq M$.

Lemma 2.21 (Zorn's Lemma)**.** *Let P be a poset. If every chain in P has an upper bound, then P has maximal elements.*

Proof Strategy: Contradiction + Ultrafilter Extension: Note that given a finite collection of sets $A_1, A_2, \ldots, A_k$ with nonempty intersection, we can build a filter $F = \{C : C \supseteq A_1 \cap A_2 \cap \cdots \cap A_k\}$ by taking **supersets** of **"minimal" elements.** This general idea comes in different forms, and it is useful in ultrafilter proofs. The global strategy is proof by contradiction, and the argumentation depends on showing how to "extend" the ultrafilter $\mathcal{U}$.

A lot of topology stuff omitted because I don't understand it. I'm pretty sure the takeaway is showing that there exists a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$ with $\mathcal{U} + \mathcal{U} = \mathcal{U}$, where $+$ is ultrafilter addition.

Let $\mathrm{FS}(X)$ be the set of finite sums of $X \subseteq \mathbb{N}$.

Theorem 2.22 (Hindman's Theorem)**.** *For any k-coloring of $\mathbb{N}$, there is an infinite $X \subseteq \mathbb{N}$ with $FS(X)$ monochromatic.*

Proof Idea. Let $\mathcal{U}$ be a free ultrafilter on $\mathbb{N}$ with $\mathcal{U} + \mathcal{U} = \mathcal{U}$. The coloring gives $\mathbb{N} = C_1 \sqcup C_2 \sqcup \cdots \sqcup C_k$. Some $C_j \in \mathcal{U}$. Let $A_0 = C_j$. We construct a chain $A_0 \supseteq A_1 \supseteq \ldots$ s.t. $A_i \in \mathcal{U}$, and elements $a_1, a_2, \ldots$ s.t. $a_i \in A_i, a_{i+1} + A_{i+1} \subseteq A_i$ for all $i \in \mathbb{N}$. $X = \{a_1, a_2, \ldots\}$ has $\mathrm{FS}(X)$ monochromatic, because any finite sum of X is in some $A_i \subseteq A_0$. $\square$

Conceptual Strategy: Generally speaking, we see a similar **iterative argument** as in past Ramsey-type theroems. However, what **propagates** this iterative argument is **not PHP**, but instead properties of **ultrafilters.**

10/09 - 10/11 Notes: Dynamical Systems

Definition 2.23. $A \subseteq \mathbb{N}$ is **syndetic** if has **bounded gaps**, i.e. $\exists m \in \mathbb{N}$ s.t. $\forall k \in \mathbb{N}$, $[k, k+m] \cap A \neq \emptyset$.

Definition 2.24. $B \subseteq \mathbb{N}$ is **thick** if it contains **arbitrarily long intervals**, i.e. $\forall p \in \mathbb{N}$, $\exists n \in \mathbb{N}$ s.t. $[n, n+p] \subseteq B$.

Fact 2.25. *$B \subseteq \mathbb{N}$ is thick iff it intersects every syndetic set.*

Definition 2.26. $S \subseteq \mathbb{N}$ is **piecewise syndetic** if it is the intersection of a syndetic set and a thick set.

Conceptually, $S \subseteq \mathbb{N}$ is piecewise syndetic if it has bounded gaps inside arbitrarily large intervals. We will show later that for any k-coloring of $\mathbb{N}$, some color class is piecewise syndetic. First, we need to understand dynamical systems.

Definition 2.27. A **dynamical system** is a pair (X, T) where X is a compact metric space and $T : X \to X$ is continuous.

Definition 2.28. (Y, T) is a **subsystem** of DS (X, T) if Y closed and nonempty, $Y \subseteq X$, and $T(Y) \subseteq Y$.

Definition 2.29. A **cyclic dynamical system** is a dynamical system where $\exists k \in \mathbb{N}$ s.t. $T^k = I$.

Definition 2.30. A point $x \in X$ is **recurrent** in a DS (X, T) if for any open set $V \ni x$, there is an $n \in \mathbb{N}$ s.t. $T^n(x) \in V$.

Definition 2.31. A point $x \in X$ is **uniformly recurrent** in a DS (X, T) if for any open set $V \ni x$, the set $\{n \in \mathbb{N} : T^n x \in V\}$ is **syndetic**.

Note: If we are considering recurrent or uniformly recurrent points in a subsystem (Y, T), then the open sets V must have $V \subseteq Y$, not $V \subseteq X$.

Fact 2.32. *Given a DS (X, T), $Y_x = \mathrm{cl}(\{T^n x : n \in \mathbb{N}\})$ is a subsystem $\forall x \in X$.*

Lemma 2.33. *Every DS (X, T) has a **minimal subsystem**, and every point in a minimal subsystem is **recurrent.***

Proof Idea. First statement follows by Zorn's Lemma. Every chain $\{Y_j | j \in J\}$ has a lower bound $\bigcap Y_j$. Let $Y_0 \subseteq X$ be a minimal subsystem, and let $x \in Y_0$ be arbitrary. I want to show x is recurrent. Consider subsystem $Y = \mathrm{cl}(\{T^n x : n \in \mathbb{N}\})$. $Y \subseteq Y_0$, so $Y = Y_0$. Thus for every open set $V \ni x$, there must be an $n \in \mathbb{N}$ s.t. $T^n x \in V$, or else $V \backslash \{x\} \notin Y$, a contradiction. $\square$

Lemma 2.34. *Every DS (X, T) has a **uniformly recurrent** point.*

Proof Idea. Let (Y, T) be a minimal subsystem of (X, T). Let $y \in Y$, $V \ni y$ open be arbitrary. I claim $\{n \in \mathbb{N} : T^n y \in V\}$ is syndetic. For each $n \in \mathbb{N}$, let $T^{-n}V = \{z \in Y : T^n z \in V\}$. We argue $Y = \bigcup_{n \in \mathbb{N}} T^{-n}V$ by minimality of Y. Thus $Y = \bigcup_{n \in F} T^{-n}V$ for F finite, because Y is compact, so every open cover has a finite subcover. Thus for all $m \in \mathbb{N}$, there is a $k \in F$ s.t. $T^k(T^m y) \in V$ giving $T^{k+m} y \in V$. This implies $\{n \in \mathbb{N} : T^n y \in V\}$ is syndetic. $\square$

Conceptual Idea: The **core ideas** are to consider a **minimal subsystem** Y; consider the **preimages** $T^{-n}V$ of V; and use **compactness** of Y to turn **infinite** cover into **finite** subcover, which ultimately gives syndetic.

We now consider k-colorings of $\mathbb{N}$ in the context of dynamical systems. Let $X = [k]^{\mathbb{N}}$ be the space of k-colorings of $\mathbb{N}$. Let $d(f, g) = (\min\{n \in \mathbb{N} : f(n) \neq g(n)\})^{-1}$. Let $T : [k]^{\mathbb{N}} \to [k]^{\mathbb{N}}$ be the "left-shift" map $(x_1, x_2, \dots) \to (x_2, x_3, \dots)$. Thus (X, d) is a compact metric space and T is a continuous map. Thus (X, T) is a dynamical system.

Theorem 2.35 (Colorings and Piecewise Syndetic)**.** *For any k-coloring of $\mathbb{N}$, some color class is **piecewise syndetic.***

Proof Idea. Let $f : \mathbb{N} \to [k]$ be arbitrary. Consider f as a point in DS (X, T). Let $Y = \text{cl}(\{T^n f : n \in \mathbb{N}\}$, so Y is a subsystem. Y has a uniformly recurrent point $h : \mathbb{N} \to [k]$. In h, all finite prefixes $h[1 : n]$ occur syndetically. If $h = T^n f$ for some $n \in \mathbb{N}$, then f starts having syndetic chunks once reaching the n'th digit (where h starts). Thus f is actually syndetic. If $h = \lim_{k\to\infty} T^{n_k} f$ for some sequence $n_1, n_2, \cdots \in \mathbb{N}$, we can still say that for an arbitrarily large $m \in \mathbb{N}$, there is some place in f (at digit n_i) where $f[n_i : (n_i + m)]$ matches $h[1 : m]$. We know $h[1]$ occurs syndetically in $h[1 : m]$, so $h[1]$ occurs syndetically in $f[n_i : (n_i + m)]$. Thus $h[1]$ is piecewise syndetic in f. □

General Strategy: The major strategy seems to be a) knowing the basics of dynamical systems, and b) interpreting them within the context of k-colorings of $\mathbb{N}$. In this sense there is not too much nontrivial creativity in the solutions.

10/21 - 10/25 Notes: Discrete Fourier Analysis

Theorem 2.36 (Szemerédi's Theorem)**.** *$\forall \delta > 0$ $\forall m \in \mathbb{N}$, $\exists n_0 \in \mathbb{N}$ s.t. $\forall n \geq n_0$ $\forall A \subseteq [n]$ s.t. $|A| \geq \delta n$, there is an m-AP in A.*

Conceptual Idea: For any positive density δ, every set A with $|A| \geq \delta n$ for large n contains an m-AP.

Theorem 2.37 (Roth's Theorem)**.** *$\forall \delta > 0$, $\exists n_0 \in \mathbb{N}$ s.t. $\forall n \geq n_0$ $\forall A \subseteq [n]$ s.t. $|A| \geq \delta n$, there is a 3-AP in A.*

Proposition 2.38. *Let $\delta > 0$, $N = N(\delta)$ large, $A \subseteq [N]$ s.t. $|A| \geq \delta N$ be arbitrary. Then i) A has a 3-AP, or ii) there is a subprogression $P \subseteq [N]$ of length N' s.t. $|A \cap P| \geq (\delta + c(\delta))|P|$ where $N' \to \infty$ as $N \to \infty$, and $c(\delta)$ is bounded away from 0 when δ is bounded away from 0.*

Conceptual Idea: For $A \subseteq [N]$ s.t. $|A| \geq \delta N$ where N is large, we can always either i) get a 3-AP, or ii) find a subprogression $P \subseteq [N]$ where A has a substantial higher density on P. If we iterate this proposition, this eventually forces that for some large N, A will have a 3-AP. On a high level it's actually similar to the color-focused idea from Theorem 2.9. We show that we get what we want, or we can keep "adding pressure."

I will just go through some basics of Discrete Fourier Analysis.

Definition 2.39. The **fourier transform** $\hat{f} : \mathbb{Z}/n \to \mathbb{C}$ of a function $f : \mathbb{Z}/n \to \mathbb{C}$ is given by $\hat{f}(r) = \mathbb{E}_x[f(x)\omega^{-rx}]$ for $\omega = e^{-2\pi i/n}$.

Fact 2.40 (Inverse DFT)**.** *For a function $f : \mathbb{Z}/n \to \mathbb{C}$, $f(x) = \sum_r \hat{f}(r)\omega^{rx}$.*

Definition 2.41. For two functions $f, g : \mathbb{Z}/n \to \mathbb{C}$,

- **Inner Product:** $\langle f, g\rangle = \mathbb{E}_x f(x)\overline{g(x)}$.
- **Norm:** $||f||_p = (\mathbb{E}_x|f(x)|^p)^{1/p}$.

- **Convolution:** $(f * g)(x) = \mathbb{E}_{y+z=x} f(y)g(z)$.

For their DFTs $\hat{f}, \hat{g}$, the expectations are replaced with sums.

Fact 2.42 (DFT Facts on f, g)**.** *For two functions $f, g : \mathbb{Z}/n \to \mathbb{C}$,*

- ***Inner Product:*** $\langle f, g\rangle = \langle \hat{f}, \hat{g}\rangle$
- ***Convolution:*** $(f \hat{*} g)(r) = \hat{f}(r) \cdot \hat{g}(r)$

3 Polynomial Methods

10/28 - 11/01 Notes: Szemerédi-Trotter Theorem

First, we talk about planar graphs. Next, we state and prove the Szemerédi-Trotter Theorem. Lastly, we talk about the multiple applications of this theorem.

Theorem 3.1 (Euler's Formula)**.** *Let G be a connected planar graph with n vertices, m edges, f faces. Then $n - m + f = 2$.*

Proof. Induction on m. If $m = n - 1$, G is a tree with 1 face, so $n - m + f = 2$. For every edge added, we add another face. □

Lemma 3.2. *All planar graphs G on n vertices have $\leq 3n - 6$ edges.*

Proof. Add up the number of edges on all faces. This is at most $2m$, because we count each edge at most twice. This is at least $3f$, since every face has at least 3 edges. Then $3f \leq 2m$ giving $f \leq (2/3)m$. Use Euler's Formula. □

Definition 3.3. The **crossing number** of a **drawing** of a graph G in $\mathbb{R}^2$ is the number of transversal intersections of the edges. The **crossing number** of a graph G is the **minimum** crossing number of all **drawings** of G.

Lemma 3.4. *$cr(G) \geq m - 3n$.*

Lemma 3.5. *$cr(G) \geq \Omega\left(\frac{m^3}{n^2}\right)$ when $m \geq 8n$.*

Proof Idea. Generate a random subgraph G' by including each vertex independently with probability p. $\mathbb{E}[\mathrm{cr}(G')] \geq \mathbb{E}[m' - 3n'] = p^2 m - 3pn$. Also, $\mathbb{E}[\mathrm{cr}(G')] \leq p^4 \mathrm{cr}(G)$ because a crossing only "survives" in G if all 4 vertices survive. Get lower bound for $\mathrm{cr}(G)$ and optimize for p. □

Conceptual Strategy: The probabilistic method not only allows us to prove **existence** of "large" combinatorial objects, but can also be useful in **bounds.** It allows us to inject an **optimization problem** w.r.t. p.

Definition 3.6. Given a set $P \subseteq \mathbb{R}^2$ of points, L a set of lines in $\mathbb{R}^2$, an **incidence** is a pair $(p, \ell) \in P \times L$ s.t. $p \in \ell$.

Note: Incidences allow for "double-counting" of points. The intersection point p of two lines ℓ_1, ℓ_2 gives **two incidences** (p, ℓ_1) and (p, ℓ_2)!

Let $I(P, L)$ be the number of incides of P, L.

Theorem 3.7 (Szemerédi-Trotter Theorem)**.** *Suppose $|P| = n, |L| = m$. Then $I(P, L) = O(n^{2/3}m^{2/3} + n + m)$.*

Proof Idea. Interpret point-line configuration as the drawing of graph G in $\mathbb{R}^2$, where incidences = vertices, edges = line segments, and we chop off the rays of each line. $I(P, L) = (\#\text{ edges}) + m$. $\mathrm{cr}(G) \leq m^2$. Use Lemma 3.5 to upper bound $I(P, L)$. ☐

Note: This is optimal. Consider $P = N \times N^2$ lattice grid. Consider L as set of lines with slope $1, 2, \ldots, N$ starting at points $1, 2, \ldots, N^2$ in first column. Easy to show $I(P, L) = \Theta(n^{2/3}m^{2/3})$. Incidences are packed very efficiently.

Definition 3.8. Given $A \subseteq \mathbb{R}$, the **sumset** is $A + A = \{a + b : a, b \in A\}$. The **productset** is $A \cdot A = \{a \cdot b : a, b \in A\}$.

$A = [n]$ has a small sumset. $A = \{2^k : k \in [n]\}$ has a small productset. These are two structurally different sets. Elekes' Theorem says that for all $A \subseteq \mathbb{R}$, either $A + A$ or $A \cdot A$ must be large.

Theorem 3.9 (Elekes Theorem)**.** *$\forall A \subseteq \mathbb{R}$, $\max(|A + A|, |A \cdot A|) \geq \Omega(|A|^{5/4})$.*

Proof Idea. Given $A \subseteq \mathbb{R}$, let $P = \{(c, d) : c \in A + A, d \in A \cdot A\}$. Let L be the set of lines $y = a(x - a')$ where $a, a' \in A$. Get lower bound $I(P, L) \geq |A|^3$ because $(a' + b, ab)$ is on $y = a(x - a')$ for all $a, a', b \in A$. Work with bounds until you obtain the result. ☐

Conceptual Strategy: P is a reasonable choice, because we want to encode $A + A, A \cdot A$. Our **ultimate goal** is to make $I(P, L)$ large. This implies n, m large, which implies $A + A, A \cdot A$ large. We want L to have a **high density** of **incidences**, which must take the form $(a + b, cd)$. We lose a degree of freedom to maximize efficiency, and consider incidences of the form $(a' + b, ab)$ to get L.

11/01 - 11/08 Notes: Polynomial Facts and Applications

Definition 3.10. $\mathrm{Poly}_D(\mathbb{F}^n)$ is the set of polynomials in n variables, with coefficients in field $\mathbb{F}$, and of total degree at most D.

Fact 3.11 (Dimension of $\mathrm{Poly}_D(\mathbb{F}^n)$)**.** *$\dim Poly_D(\mathbb{F}^n) = \binom{D+n}{n}$ using stars and bars with the monomial basis.*

Fact 3.12 (Vanishing Sets $S \subseteq \mathbb{F}^n$)**.** *If $S \subseteq \mathbb{F}^n$ has $|S| < \dim Poly_D(\mathbb{F}^n)$, there exists a nonzero $p \in Poly_D(\mathbb{F}^n)$ vanishing on S. Inversely for any finite $S \subseteq \mathbb{F}^n$, there exists a nonzero p vanishing on S with degree $\leq n|S|^{1/n}$.*

Fact 3.13 (High Vanishing $\implies$ Zero Polynomial)**.** *If $p \in Poly_D(\mathbb{F})$ vanishes at $D+1$ points, then $p = 0$. If $p \in Poly_D(\mathbb{F}^n)$ vanishes at $D+1$ points on a line ℓ, then p vanishes on the entire line ℓ. If $p \in Poly_{q-1}(\mathbb{F}_q^n)$ vanishes on $\mathbb{F}_q^n$, then $p = 0$.*

Fact 3.14 (Division Theorem in One Variable)**.** *If $p \in Poly_D(\mathbb{F})$ and $x_1 \in \mathbb{F}$, then $p(x) = (x - x_1)p_1(x) + r$ where $p_1 \in Poly_{D-1}(\mathbb{F}), r \in \mathbb{F}$.*

Definition 3.15. A **joint** of L is a point contained in three lines not coplanar.

Let $J(L)$ be the number of joints of L.

Lemma 3.16. *Some line contains $\leq 3J^{1/3}$ joints.*

Proof. Let p be the lowest-degree nonzero polynomial vanishing on all joints of L. p is a polynomial over $\mathbb{R}^3$ and vanishes on $J \subseteq \mathbb{R}^3$. We know p is low-degree in the sense that $\deg(p) \leq 3J^{1/3}$. If there are $> 3J^{1/3}$ joints on every line ℓ, then p vanishes on all lines ℓ. Then ∇p vanishes on all joints J, a contradiction. □

Conceptual Strategy: Understand the direct relationship between **vanishing sets** and **polynomial degrees.** Once starting contradiction world, **pump** as much **vanishing** as possible into the "low-degree" polynomial.

Definition 3.17. A **line** $L \subseteq \mathbb{F}_q^n$ is $L = \{at + b : t \in \mathbb{F}_q\}$ for $a, b \in \mathbb{F}_q^n$.

Definition 3.18. $N \subseteq \mathbb{F}_q^n$ is a **Nikodym set** if $\forall x \in \mathbb{F}_q^n$, there is a line $L(x)$ containing x s.t. $L(x) \backslash \{x\} \subseteq N$.

Theorem 3.19 (Dvir)**.** *All Nikodym sets $N \subseteq \mathbb{F}_q^n$ have size at least $c_n \cdot q^n$.*

Proof. Let $N \subseteq \mathbb{F}_q^n$ be a Nikodym set. AFSOC $|N| < \frac{1}{(10n)^n} q^n$. Let p be the lowest-degree nonzero polynomial vanishing on N. We know p is low-degree in the sense that $\deg(p) \leq n|N|^{1/n} < \frac{1}{10}q < q - 1$. But $\forall x \in \mathbb{F}_q^n$, p vanishes on $L(x)\backslash\{x\}$ which has size $q-1$, so p vanishes on the entire line $L(x)$, so p vanishes on x. Thus p vanishes on $\mathbb{F}_q^n$, but p has $\deg(p) < q - 1$, so $p = 0$, a contradiction. □

Conceptual Strategy: Same as before. If N is small, then p is low-degree, but p has high vanishing (via Nikodym condition), a contradiction.

Definition 3.20. $K \subseteq \mathbb{F}_q^n$ is a **Kakeya set** if it contains a line in every direction, i.e. $\forall a \in \mathbb{F}_q^n \backslash \{0\}$, $\exists b \in \mathbb{F}_q^n$ s.t. $L = \{at + b : t \in \mathbb{F}_q\} \subseteq K$.

Theorem 3.21 (Dvir)**.** *All Kakeya sets $K \subseteq \mathbb{F}_q^n$ have size at least $c_n \cdot q^n$.*

Proof Idea. Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set. AFSOC $|K| < \frac{1}{(10n)^n} q^n$. Let p be the lowest-degree nonzero polynomial vanishing on K. We know p is low-degree in the sense that $\deg(p) \leq n|K|^{1/n} < \frac{1}{10}q < q - 1$. Let $D = \deg(p)$, and write $p = p_D + Q$ where p_D is the degree-D part of p. We know $\forall a \in \mathbb{F}_q^n \backslash \{0\}$, $\exists b \in \mathbb{F}_q^n$ s.t. p vanishes on line $at + b$, which has q points. Thus p is the zero polynomial when restricted to the line, i.e. $R(t) = p(at+b)$ has $R(t) = 0$. Thus $p(at+b) = 0$ giving $p_D(at+b) + Q(at+b) = 0$. Focusing on the coefficient of t^D, this implies $p_D(a) = 0$. Thus $\forall a \in \mathbb{F}_q^n \backslash \{0\}$, $p_D(a) = 0$, so $p_D = 0$, a contradiction. □

Conceptual Strategy: Same as before. However, the **contradiction strategy** was to **violate** the **minimality** of p, which seems useful in future contexts.

Definition 3.22. $C \subseteq \mathbb{F}_3^n$ is a **capset** if for distinct $x, y, z \in C$, $x + y + z \neq 0$.

Capsets and Slice-Rank Method omitted because I don't understand it. Also it was omitted from Florian's topic list.

11/08 - 11/08 Notes: Combinatorial Nullstellensatz

Theorem 3.23 (Combinatorial Nullstellensatz)**.** *Let $p \in Poly_D(\mathbb{F}^n)$ with a nonzero ${x_1}^{d_1}{x_2}^{d_2}\ldots{x_n}^{d_n}$-coefficient where $\sum d_i = D$. Then p does not vanish on any set $E_1 \times E_2 \times \cdots \times E_n \subseteq \mathbb{F}^n$ where $|E_i| > d_i$ for all $i \in [n]$.*

Conceptual Strategy: Theorem 3.23 is actually useful for the following contradiction strategy. Get a polynomial p that has a low "max-degree" term ${x_1}^{d_1}{x_2}^{d_2}\ldots{x_n}^{d_n}$ and vanishes on a "large" set. This gives a contradiction.

Proof Idea. Induction on $D = \sum d_i$. Use Division Theorem. □

Theorem 3.24 (Cauchy-Davenport Theorem)**.** *Let p be prime. Then for all nonempty $A, B \subseteq \mathbb{F}_p$, $|A + B| \geq \min\{|A| + |B| - 1, p\}$.*

Proof. If $|A| + |B| > p$, then $A \cap (x - B) \neq \emptyset$ for all $x \in \mathbb{F}_p$, so $A + B = \mathbb{F}_p$. Otherwise $|A| + |B| \leq p$. AFSOC $A + B \subseteq C$ for $|C| = |A| + |B| - 2$. Consider $p(x, y) = \prod_{c \in C}(x+y-c)$. p is low-degree in the sense that $\deg(p) = |A|+|B|-2$, and vanishes on a large set $A \times B$. p has nonzero max-degree term $x^{|A|-1}y^{|B|-1}$, giving the contradiction. □

Conceptual Idea: Same conceptual strategy as described. To construct p, remember we want p to be "low-degree" and "high-vanishing." In contradiction world, $|C|$ seems to be "small", so our polynomial p should probably have low degree $|C|$. Further, we want p to be "high-vanishing", which motivates our specific choice.

11/11 - 11/15 Notes: More Polynomial Theorems

Lemma 3.25. *If $\mathbb{F}$ finite field, then $\sum_{x \in \mathbb{F}} x^i = 0$ for $i < |\mathbb{F}| - 1$.*

Proof. $x^i - 1$ has at most i zeros, so $\exists a \in \mathbb{F}\backslash\{0\}$ with $a^i \neq 1$.Then $\sum_x x^i = \sum_x (ax)^i = a^i \sum_x x^i$ giving $\sum_x x^i = 0$. □

Lemma 3.26 (Looping Lemma)**.** *$\mathbb{F}$ finite field. Then $\forall p \in Poly_D(\mathbb{F}^n)$ s.t. $D < n(|\mathbb{F}| - 1)$, $\sum_{x \in \mathbb{F}^n} p(x) = 0$.*

Conceptual Idea: If p is low-degree, then $\sum_{x \in \mathbb{F}^n} p(x) = 0$.

Lemma 3.27 (Fermat's Little Theorem)**.** *$\forall x \in \mathbb{N}$, $x^p \equiv x \mod p$ for p prime.*

Note: Lemma 3.27 is useful in constructing **indicators** from **polynomials** in $\mathbb{F}_p$. The function $f(a) = a^{p-1} \mod p$ equals 1 if $a \neq 0$, and 0 if $a = 0$. This can be useful for a) logic or b) counting.

Theorem 3.28 (Chevalley-Warning Theorem)**.** *Let p be prime. Then for all polynomials $p_1, p_2, \ldots, p_k \in Poly(\mathbb{F}^n)$ s.t. $\deg(p_1) + \deg(p_2) + \cdots + \deg(p_k) < n$, the number of common zeros of $p_1, p_2, \ldots, p_k$ is divisible by p.*

Proof. Let $P(x) = \prod_{i=1}^{k} \left(1 - (p_i(x))^{p-1}\right)$. $P(x) = 1$ if $p_i(x) = 0$ for all $i \in [k]$, 0 otherwise. $\deg(P) = (p-1)\sum \deg(p_i) < n(p-1)$. Thus $\sum_{x \in \mathbb{F}^n} P(x) = 0$. Thus the number of times $P(x) = 1$ is divisible by p, so the number of common zeros of $p_1, p_2, \ldots, p_k$ is divisible by p. □

Conceptual Strategy: Theorem 3.28 is actually useful in the following way. Suppose we have polynomials $p_1, p_2, \ldots, p_k$ with low total degree and a common zero (often $x = 0$). Then they share another common zero.

Theorem 3.29 (Erdös-Ginzburg-Ziv Theorem)**.** *Let $a_1, a_2, \ldots, a_{2n-1} \in \mathbb{Z}/n$ be arbitrary. Then there is a **zero-sum** subsequence of **length n.***

Note: This is optimal. Consider $n-1$ 0's and $n-1$ 1's.

Proof Idea. First, consider $n = p$ prime. Let $p_1(x_1, x_2, \ldots, x_{2p-1}) = \sum a_i x_i^{p-1}$ and $p_2(x_1, x_2, \ldots, x_{2p-1}) = \sum x_i^{p-1}$. $p_1(0) = p_2(0) = 0$ and p_1, p_2 have low total degree in the sense that $\deg(p_1) + \deg(p_2) = 2p - 2 < 2p - 1$. Thus by Chevalley-Warning Theorem, there is another common zero $x \in \mathbb{F}_p^{2p-1} \backslash \{0\}$, meaning $p_1(x) = p_2(x) = 0$. Let $I = \{i \in [2p-1] : x_i \neq 0\}$ be the set of nonzero components of x. $p_2(x) = 0$ implies I has size p. $p_1(x) = 0$ implies $\sum_{i \in I} a_i = 0$.

Next, we show that if the claim holds for k, m then it holds for $n = km$. □

Theorem 3.30 (Frankl-Wilson Theorem)**.** *Let $n = 4p$ for p prime. Let $F \subseteq \{-1, 1\}^n$ s.t. no two vectors in F are orthogonal. Then $|F| \leq 4\left(\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{p-1}\right)$.*

4 Geometric Combinatorics

Q: Why are polynomials everywhere in combinatorics?

- **A1: Deletion-Contraction Formulas** for combinatorial objects.
- **A2: Ehrhart Theory** involves counting integer vectors in a polytope.

11/15 - 11/18 Notes: Deletion-Contraction Formulas

Definition 4.1. The **chromatic polynomial** $\chi_G : \mathbb{N} \to \mathbb{N}$ of a finite graph G is $\chi_G(k) =$ number of proper k-colorings of G.

Theorem 4.2 (Chromatic Polynomial)**.** *For all graphs G, χ_G is polynomial of degree n.*

Proof. Let $e \in E$ be arbitrary. Let $G \backslash e$ be "G delete e" and G/e be "G contract e". Argue $\chi_G(k) = \chi_{G\backslash e}(k) - \chi_{G/e}(k)$. Induction on m. □

Definition 4.3. An **acyclic orientation** of a graph $G = (V, E)$ is a choice of direction for each edge $e \in E$ s.t. there are **no directed cycles.**

Lemma 4.4. $|\chi_G(-1)|$ *is the number of ACs of* G.

Proof. Let $A(G)$ be the number of ACs of G. Let $e \in E$ be arbitrary. Argue $A(G) = A(G\backslash e) + A(G/e)$. Given an AC of $G\backslash e$, there is at least one way to orient e, to get an AC of G. If both ways work, then there is an AC of G/e. This induces a bijection between ACs of G and ACs of $G\backslash e, G/e$. Thus $\chi_G(-1)$ and $A(G)$ follow essentially the same recursive formula (up to an overall sign), and match at base cases, so $A(G) = |\chi_G(-1)|$. □

11/18 - 11/25 Notes: Ehrhart Polynomial and Applications

Theorem 4.5 (Pick's Theorem)**.** *Let* $P \subseteq \mathbb{R}^2$ *be a lattice polygon. Then* $Area(P) = i + b/2 - 1$*, where* $i = \#$ *interior lattice points and* $b = \#$ *boundary lattice points.*

Proof Idea. If theorem holds for two of $P, Q, P \cup Q$ where P, Q lattice polygons with disjoint interiors, it holds for the third. We then show theorem holds for axis-aligned rectangles, then axis-aligned right triangles, then lattice triangles, then lattice polygons. □

Definition 4.6. A **lattice polytope** $P \subseteq \mathbb{R}^d$ is the convex hull of finitely many points in $\mathbb{Z}^d$.

Theorem 4.7 (Ehrhart's Theorem)**.** *Let* $P \subseteq \mathbb{R}^d$ *be a full-dimensional lattice polytope. Define* $L : \mathbb{Z}_{\geq 0} \to \mathbb{R}$ *s.t.* $L(t) = |tP \cap \mathbb{Z}^d|$ *for all* $t \in \mathbb{Z}_{\geq 0}$*. Then* L *is a polynomial of degree* d *with leading coefficient of* $vol(P)$.

Note: I'm pretty sure Theorem 4.7 holds for non-full-dimensional lattice polytopes P of dimension $j < d$. In the j-dimensional subspace of $\mathbb{R}^d$ containing P, P is full-dimensional, and this subspace is isomorphic to $\mathbb{R}^j$. I'm pretty sure this means L will be degree j, but the leading coefficient may not be vol(P), because the lattice could get wonky under the transformation from the subspace to $\mathbb{R}^j$.

Note: L is counting **all** lattice points of P, meaning both **interior points** and **boundary points.**

Lemma 4.8 (Inclusion-Exclusion and Polynomials)**.** *For a function* $f : \mathbb{Z}_{\geq 0} \to \mathbb{R}$ *and* $d \in \mathbb{Z}_{\geq 0}$*, the following are equivalent.*

1. ***Inclusion-Exclusion:*** $\forall t \in \mathbb{Z}_{\geq 0}$, $\sum_{k=0}^{d+1} (-1)^{d+1-k} \binom{d+1}{k} f(t+k) = 0$.
2. ***Polynomial:*** f *is a polynomial of degree* $\leq d$.

Proof Idea for Theorem 4.7. Use Lemma 4.8. Argue WLOG P is a lattice simplex, meaning P is the convex hull of $d+1$ points in $\mathbb{R}^d$. Show L satisfies the Inclusion-Exclusion Formula of Lemma 4.8. □

Theorem 4.9 (Ehrhart-MacDonald Reciprocity)**.** $(-1)^d L(-t) = |t\mathring{P} \cap \mathbb{Z}^d|$.

We now discuss some applications of Ehrhart's Theorem.

Lemma 4.10 (Integer Flows)**.** *Let G be a directed graph with source s, sink t. Let $F(C)$ be the number of integer flows with capacity C in G. Then $F(C)$ is a polynomial in C.*

Proof. The set of flows of capacity 1 is a rational polytope $P \subseteq \mathbb{R}^m$. The set of flows of capacity C is the rational polytope $C \cdot P \subseteq \mathbb{R}^m$. The number of integer flows of capacity C is the number of lattice points in $C \cdot P$. Via Theorem 4.7, this is a polynomial in C. □

Lemma 4.11 (Chromatic Polynomial)**.** *For all graphs G, χ_G is a polynomial of degree n.*

Proof. Let $P = [0,1]^n$. $kP = [0,k]^n$ is the polytope of $(k+1)$-colorings of G, and contains $(k+1)^n$ lattice points. To count proper $(k+1)$-colorings of G, each edge $(i,j) \in E$ removes the hyperplane $x_i = x_j$ from P. But the number of lattice points on these hyperplanes are polynomials of degree $n-1$. Thus the final number of lattice points, the proper $(k+1)$-colorings of G, will be a polynomial of degree n. □

In the simplex $\Delta_d = \{x \in \mathbb{R}^d : 0 \le x_1 \le x_2 \le \cdots \le x_d \le 1\}$, we can count lattice points in $n \cdot \Delta_d$ (multisets of values in $[0,n]$) and lattice points in $n \cdot \mathring{\Delta}_d$ (sets of values in $[0,n]$) using the same polynomial $L(t)$! Of course, this can be done using stars and bars, but shows an interesting connection.

Lastly, we can associate lattice points $x = (m_1, m_2, \ldots, m_d) \in \mathbb{Z}^d$ with monomials $z = z_1^{m_1} z_2^{m_2} \ldots z_d^{m_d}$, and sets of lattice points with the formal sum of the monomials.

12/02 - 12/04 Notes: Ehrhart Series and More Applications

Definition 4.12. The **generating function** of $S \subseteq \mathbb{R}^d$ is $G_s(z) = \sum_{\alpha \in S \cap \mathbb{Z}^d} z^\alpha$, where $z^\alpha = {z_1}^{\alpha_1} z_2^{\alpha_2} \ldots z_d^{\alpha_d}$.

Definition 4.13. Let $w_1, w_2, \ldots, w_d \in \mathbb{Z}^d$ be a basis of $\mathbb{R}^d$. $K = \{\lambda_1 w_1 + \lambda_2 w_2 + \cdots + \lambda_d w_d : \lambda_i \ge 0 \ \forall i \in [d]\}$ is the **simplical cone** of the w_i's. $\Pi = \{\lambda_1 w_1 + \lambda_2 w_2 + \cdots + \lambda_d w_d : 0 \le \lambda_i < 1 \ \forall i \in [d]\}$ is the **fundamental parallelepiped** of the w_i's.

Theorem 4.14 (GF for Cones and Fund. Ppds)**.** *$\forall v \in \mathbb{R}^d$ and $\forall w_1, w_2, \ldots, w_d \in \mathbb{R}^d$ basis,*

$$G_{v+K}(z) = \frac{G_{v+\Pi}(z)}{(1-z^{w_1})(1-z^{w_2})\cdots(1-z^{w_d})}$$

Theorem 4.15 (Ehrhart Series)**.** *Let $P = \operatorname{conv}(v_1, v_2, \ldots, v_{d+1}) \subseteq \mathbb{R}^d$ be a lattice simplex. Let $w_i = (v_i, 1) \in \mathbb{R}^{d+1}$ for each $i \in [d+1]$. Let $\Pi \subseteq \mathbb{R}^{d+1}$ be the fund. ppd. of the w_i's. Then $E_P(z) = 1 + \sum_{t=1}^{\infty} L_P(t)z^t$ has*

$$E_P(z) = \frac{h_d^* z^d + h_{d-1}^* z^{d-1} + \cdots + h_0^*}{(1-z)^{d+1}}$$

giving

$$L_p(t) = h_0^* \binom{t+d}{d} + h_1^* \binom{t+d-1}{d} + \cdots + h_d^* \binom{t}{d}$$

where h_k^* is the number of lattice points in $\Pi \subseteq \mathbb{R}^{d+1}$ at level k (i.e. $x_{d+1} = k$) for all $k \in [d+1]$.

Conceptual Strategy: This allows us to compute the Ehrhart Polynomial $L_P(t)$ (in the binomial basis) by computing $h_0^*, h_1^*, \ldots, h_d^*$, which could be computationally easier.

Let's now discuss more applications.

Theorem 4.16 (Euler Relation)**.** *Let $P \subseteq \mathbb{R}^d$ be a full-dimensional lattice polytope. Let f_j be the number of faces of dimension $j \in [0, d]$. Then $\sum_{j=0}^{d} (-1)^j f_j = 1$.*

Proof. $L_P(t) = \sum_{F \text{ face of } P} |t\mathring{F} \cap \mathbb{Z}^d| = \sum_{F \text{ face of } P} (-1)^{\dim F} L_F(-t)$ using EM-Reciprocity. $t = 0$ gives $L(0) = 1$ for all Ehrhart polynomials, so $\sum_{j=0}^{d} (-1)^j f_j = 1$. $\square$

Definition 4.17. A **zonotope** is a polytope that is the Minkowski sum of finite closed line segments.

Theorem 4.18 (Zonotopes and Forests)**.** *Let G be a simple graph on n vertices. Let $e_1, e_2, \ldots, e_n \in \mathbb{R}^n$ be the standard basis. Define the zonotope $Z_G = \sum_{(i,j) \in E} [e_i, e_j]$ using Minkowski sum. Then $L_{Z_G}(t) = c_{n-1}t^{n-1} + c_{n-2}t^{n-2} + \cdots + c_0$, where c_k is the number of forests in G on k edges for each $k \in [0, n-1]$.*

Definition 4.19. $Z_{K_n} = \sum_{i,j \in [n]} [e_i, e_j]$ is the **permutohedron** in $\mathbb{R}^n$. It is called this because $Z_{K_n} = \operatorname{conv}\{(\pi(1) - 1, \pi(2) - 1, \ldots, \pi(n) - 1) | \pi : [n] \to [n] \text{ permutation}\}$.

12/06 - 12/06 Notes: Geometry of Numbers

Theorem 4.20 (Minkowski's Theorem)**.** *Let A be an invertible $n \times n$ matrix. Let $L = A[\mathbb{Z}^n]$ and $d(L) = |\det(A)|$. Let $C \subseteq \mathbb{R}^n$ be convex, compact, centrally symmetric. If $vol(C) \geq 2^n d(L)$, then C contains a nonzero lattice point in L.*

Note: This theorem is optimal. Consider $C \subseteq A\left((-1,1)^n\right)$. Then $\operatorname{vol}(C) < 2^n d(L)$, but can be made arbitrarily close.

Theorem 4.21 (Two Squares Theorem)**.** *Let p be prime, $p \equiv 1 \mod 4$. Then $p = x^2 + y^2$ for $x, y \in \mathbb{Z}$.*

Proof Idea. We want a large C, small A, and nonzero lattice point $(x, y) \in C$ implies $x^2 + y^2 = p$. Consider $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 2p\}$. We want to choose A s.t. $\mathrm{vol}(C) > 4d(L) \implies 2\pi p > 4|\det(A)| \implies |\det(A)| < (\pi/2)p$, and further for all $(u, v) \in \mathbb{Z}^2$, $A(u, v) = (x, y)$ with $x^2 + y^2 \equiv 0 \mod p$. We can guess A contains p as an entry, and in order to keep the determinant low, also contains 0. Testing some ideas, we land on $A = \begin{pmatrix} 1 & 0 \\ u & p \end{pmatrix}$, where u has order 4 (so $u^2 \equiv -1 \mod p$) in the multiplicative group $\mathbb{Z}_p^*$. □

Conceptual Strategy: The strategy is pretty much straightforward based on the complete statement of Theorem 4.20. The statement of the theorem shows **existence** of a **"special" lattice point**, which motivates its usage.

Theorem 4.22 (Four Squares Theorem)**.** *$\forall n \in \mathbb{N}$, n is the sum of 4 squares.*

Proof Idea. Induction gives it's enough to show statement holds for primes. Same strategy as before. We want a large C, small A, and nonzero lattice point $(w, x, y, z) \in C$ implies $w^2 + x^2 + y^2 + z^2 = p$. Consider $C = \{(w, x, y, z) \in \mathbb{R}^4 : w^2 + x^2 + y^2 + z^2 < 2p\}$. We want to choose A s.t. for all $(s, t, u, v) \in \mathbb{Z}^4$, $A(s, t, u, v) = (w, x, y, z)$ with $w^2 + x^2 + y^2 + z^2 \equiv 0 \mod p$. Testing some ideas and using some basic assumptions on A (A could be upper triangular, mostly have p's, 1's on the diagonal) leads to $A = \begin{pmatrix} p & 0 & r & s \\ 0 & p & s & -r \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ where $r^2 + s^2 + 1 \equiv 0 \mod p$. □